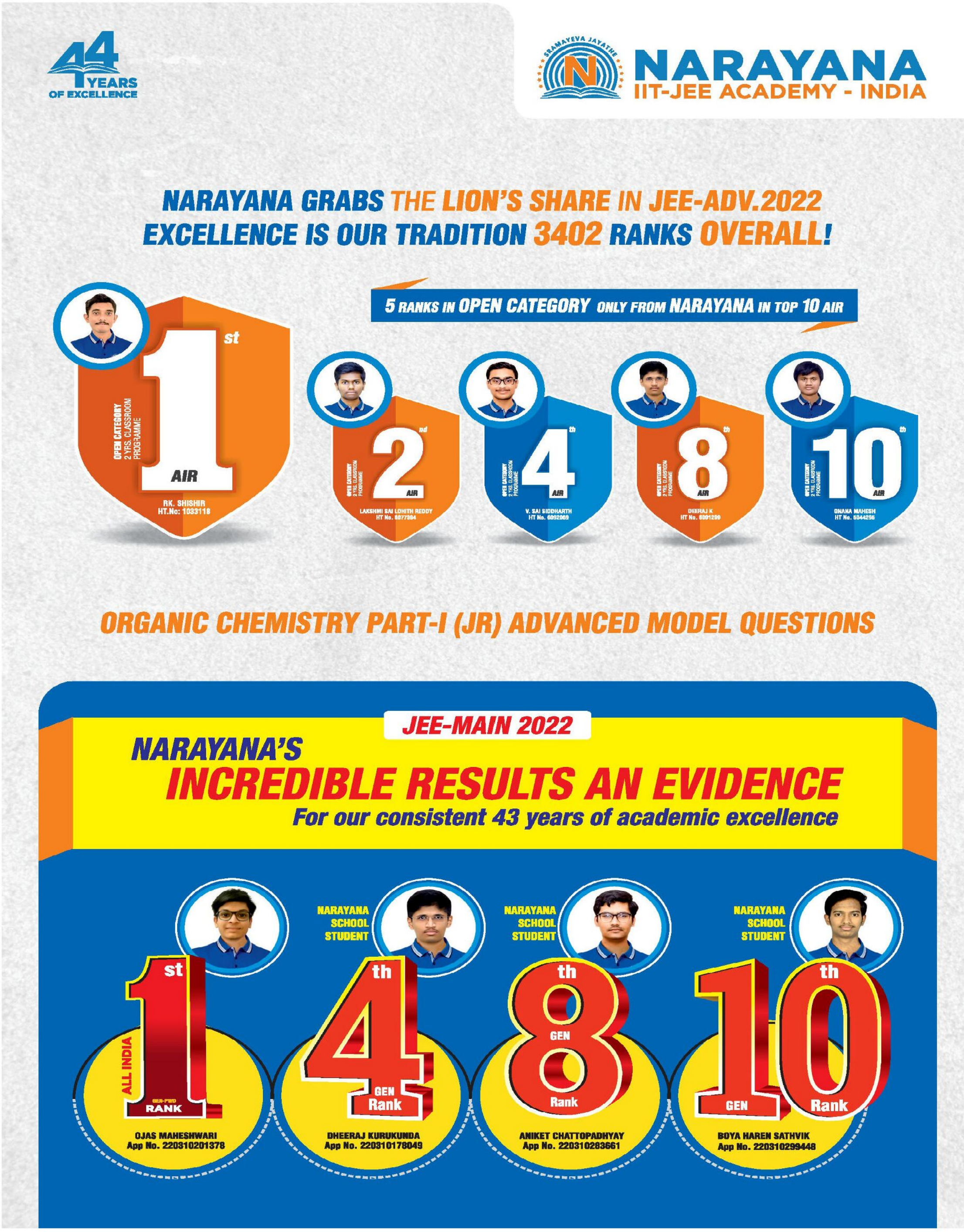


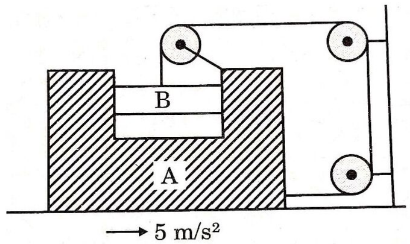
PHYSICS

MECHANICS

JEE ADVANCED IMPORTANT QUESTIONS

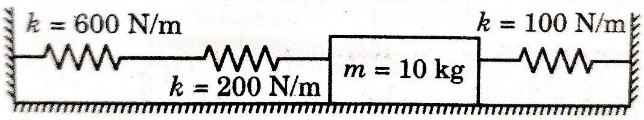
(Single Correct Type)

1. If block A is moving with an acceleration of 5 m/s^2, the acceleration of B w.r.t. ground is



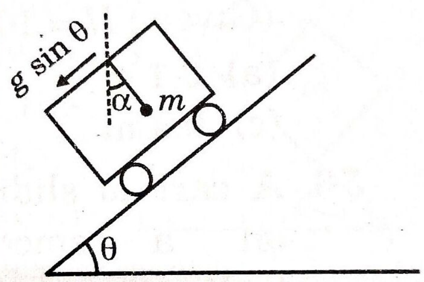
A) $5m/s^2$ B) $5\sqrt{2}\,m/s^2$ C) $5\sqrt{5}\,m/s^2$ D) $10m/s^2$

2. a 10 kg mass is stationary in the position shown. Each spring has a characteristic stiffness constant, k, as shown in the diagram. If the mass is now displaced 20 mm to the right, what will be the net force, in newtons, exerted by the three springs on the mass? Ignore gravity



A) 1.2 N B) 5 N C) 14 N D) 18 N

3. A trolley is accelerating down an incline of angle θ with acceleration g $\sin\theta$. Which of the following is correct. (α is the angle made by the string with vertical)



A) $\alpha = \theta$ B) $\alpha = 0^0$

C) Tension in the string, $T = mg$ D) Tension in the string, $T = mg\sec\theta$

4. A particle of mass m attached to a horizontal string describes a circle of radius r on a rough horizontal table. After completing one full trip around the circle the speed of particle is halved. What is the coefficient of friction in terms of mass m, radius r and initial velocity u_0?

A) $\dfrac{3u_0^2}{16\pi gr}$ B) $\dfrac{2u_0^2}{9\pi gr}$ C) $\dfrac{9u_0^2}{16\pi gr}$ D) $\dfrac{5u_0^2}{16\pi gr}$

5. A car moving at a speed of 36 km/hr is taking a turn on a circular road of radius 50 m. A small wooden plate is kept on the seat with its plane perpendicular to the radius of the circular road (fig). A small block of mass 100g is kept on the seat which rests against the plate. The friction coefficient between the block and the plate is μ = 0.58. The plate is slowly turned so that the angle between the normal to the plate and the radius of the road slowly increases. Find the angle at which the block will just start sliding on the plate